

LOCAL $\aleph_1$ -COPRESENTABILITY OF THE CATEGORY OF COMPACT HAUSDORFF SPACES

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Our purpose here is to gather several relevant results about the category **CompHaus** of compact Hausdorff spaces, and provide accessible (sic) proofs of these facts leading up to a proof that it is locally $\aleph_1$ -copresentable.

A good overview of some of these results is contained in the introduction of [3].

We first prove a couple preliminary results.

Lemma 1. *Let $\mathcal{I}$ be a cofiltered category, and let $X : \mathcal{I} \rightarrow \mathbf{CompHaus}$ be a cofiltered diagram in which X_i is non-empty for each $i \in \mathcal{I}$. Then $\lim_i X_i$ is also non-empty.*

Proof. Consider the product space $\prod_{i \in \mathcal{I}} X_i$. Now for each morphism $f : i \rightarrow j$ in $\mathcal{I}$, define the subset

$$E_f := \{x \in \prod_{i \in \mathcal{I}} X_i \mid X_f(x_i) = x_j\}.$$

Then each E_f is a closed subset.

Next, we prove that the collection $\{E_f : f \in \text{Mor}(\mathcal{I})\}$ has the finite intersection property, i.e. that $\bigcap_{f \in F} E_f$ is non-empty for every finite set $F \subseteq \text{Mor}(\mathcal{I})$. For $f \in F$ we write $f : i_f \rightarrow j_f$. Then the diagram with objects $J := \{i_f \mid f \in F\} \cup \{j_f \mid f \in F\}$ and morphisms $\{f \mid f \in F\}$ has a cone with vertex $k \in \mathcal{I}$ and morphisms $g_i : k \rightarrow i$ for each $i \in J$. Now choose $y \in X_k$, and define $x \in \prod_{i \in \mathcal{I}} X_i$ such that $x_i = X_{g_i}(y)$ if $i \in J$, with arbitrary choices of $x_i \in X_i$ for all other i . We then see that $x \in \bigcap_{f \in F} E_f$, which finishes the proof of the claim.

Since $\prod_{i \in \mathcal{I}} X_i$ is compact, that implies that the intersection of all E_f is non-empty. But that intersection is precisely $\lim_{i \in \mathcal{I}} X_i$. $\square$

Lemma 2. *Suppose we have a cofiltered limit $X = \lim_{i \in \mathcal{I}} X_i$ in **Top**. Note the topology on X is the weak topology for the projections $p_i : X \rightarrow X_i$. Then the canonical subbasis of this topology on X is closed under finite intersections. Thus, it agrees with the canonical basis of the topology on X .*

Proof. Suppose we have a finite collection of subbasic open sets of the form $U_n = p_{i_n}^{-1}(V_n)$, $n \in \{1, \dots, N\}$, where each V_n is an open subset of X_{i_n} . Take a cone $(j, f_n : j \rightarrow i_n)$ of the objects $i_1, \dots, i_N$. We then have

$$\bigcap_{n=1}^N U_n = p_j^{-1} \left(\bigcap_{n=1}^N X_{f_n}^{-1}(V_n) \right),$$

where the right hand side is again in the canonical subbasis. $\square$

Proposition 3. *The functor $\text{Hom}(-, [0, 1]) : \mathbf{CompHaus}^{\text{op}} \rightarrow \mathbf{Set}$ is monadic. (Originally proved in [1])*

Proof. We use the crude monadicity theorem (see e.g. [6], Thm. IV.4.2). First, the functor has a left adjoint $S \mapsto [0, 1]^S$ with the evident isomorphism

$$\text{Hom}_{\mathbf{CompHaus}}(X, [0, 1]^S) \simeq \text{Hom}_{\mathbf{Set}}(S, \text{Hom}_{\mathbf{CompHaus}}(X, [0, 1])).$$

To see the functor is conservative, suppose we have a continuous function $f : X \rightarrow Y$ such that $f^* : \text{Hom}(Y, [0, 1]) \rightarrow \text{Hom}(X, [0, 1])$ is a bijection. Then for any $x_1, x_2 \in X$ with $x_1 \neq x_2$,

there exists $\varphi : X \rightarrow [0, 1]$ with $\varphi(x_1) = 0$ and $\varphi(x_2) = 1$ by Urysohn's lemma. Since f^* is surjective, there exists $\psi : Y \rightarrow [0, 1]$ with $\varphi = \psi \circ f$; thus, we must have $f(x_1) \neq f(x_2)$. Likewise, we know the image of f is closed. If this image is not all of Y , then by Urysohn's lemma there exists nonzero $\varphi : Y \rightarrow [0, 1]$ which is zero on the image. But then $\varphi \circ f = 0 \circ f$, contradicting the injectivity of f^* . Thus, f is a bijective continuous function, and therefore a homeomorphism.

Finally, suppose we have a coreflexive equalizer pair

$$E \xrightarrow{i} A \begin{matrix} \xrightarrow{f} \\ \xrightarrow{g} \end{matrix} B$$

with $r : B \rightarrow A$. We may assume that i is a subspace inclusion map. We may use r to think of B as a bundle of compact spaces over A , with two sections f, g . We then need to show that

$$\mathrm{Hom}(B, [0, 1]) \begin{matrix} \xrightarrow{f^*} \\ \xrightarrow{g^*} \end{matrix} \mathrm{Hom}(A, [0, 1]) \xrightarrow{i^*} \mathrm{Hom}(E, [0, 1])$$

is a coequalizer diagram. We first define $s : \mathrm{Hom}(E, [0, 1]) \rightarrow \mathrm{Hom}(A, [0, 1])$ by choosing a Tietze extension of each continuous function $E \rightarrow [0, 1]$. Now, for each $\varphi \in \mathrm{Hom}(A, [0, 1])$, we can define a continuous function on $\mathrm{im}(f) \cup \mathrm{im}(g) \subseteq B$ to be $\varphi \circ r$ on $\mathrm{im}(f)$, and $s(i^*(\varphi)) \circ r$ on $\mathrm{im}(g)$. Note that on the overlap $\mathrm{im}(f) \cap \mathrm{im}(g) = f(E) = g(E)$, the first expression gives $f(e) \mapsto \varphi(e)$, and the second expression gives $g(e) \mapsto s(i^*(\varphi))(e) = \varphi(e)$, so we have indeed given a well-defined function on $\mathrm{im}(f) \cup \mathrm{im}(g)$. Choosing a Tietze extension of this function to a function $B \rightarrow [0, 1]$ for each φ , we get a map $t : \mathrm{Hom}(A, [0, 1]) \rightarrow \mathrm{Hom}(B, [0, 1])$. By construction, we have $i^*s = \mathrm{id}$, $f^*t = \mathrm{id}$, and $g^*t = si^*$, so we have shown that the diagram above is a split coequalizer. $\square$

This shows that $\mathbf{CompHaus}^{\mathrm{op}}$ is equivalent to the category of algebras over the monad $S \mapsto \mathrm{Hom}_{\mathbf{CompHaus}}([0, 1]^S, [0, 1])$. We may view such algebras as being models of the infinitary algebraic theory of all continuous functions $[0, 1]^S \rightarrow [0, 1]$. In fact, we can show that any such function only depends on countably many coordinates in the domain, so that operations of this theory will be generated by the continuous functions $[0, 1]^\omega \rightarrow [0, 1]$. Indeed, we get the following somewhat stronger result:

Proposition 4. *The object $[0, 1]$ of $\mathbf{CompHaus}$ is $\aleph_1$ -copresentable. (Originally proved in [2], 6.5(c))*

Proof. Suppose we have an $\aleph_1$ -cofiltered limit $X = \lim_{i \in \mathcal{I}} X_i$ with projections $p_i : X \rightarrow X_i$, and a continuous function $\varphi : X \rightarrow [0, 1]$. For the time being, fix $n \in \mathbb{N}_{>0}$. Then for any $x \in X$, there exists an interval neighborhood N_x of $\varphi(x)$ of diameter at most $1/n$ — for example, we can take $N_x := (\varphi(x) - 1/(2n), \varphi(x) + 1/(2n)) \cap [0, 1]$. We can also take a basic open neighborhood whose image is contained in N_x ; by lemma 2, we can write that basic open neighborhood in the form $p_i^{-1}(V)$ where V is an open subset of X_i .

By compactness of X , we may take finitely many such basic open neighborhoods of the form $p_i^{-1}(V)$ which cover X . Again using the assumption that $\mathcal{I}$ is cofiltered, we may assume that i is the same for each neighborhood. In particular, we see that whenever we have $x, y \in X$ with $p_i(x) = p_i(y)$, then $|\varphi(x) - \varphi(y)| < 1/n$.

To summarize, we have shown that for each $n \in \mathbb{N}_{>0}$, there exists $i \in \mathcal{I}$ and a finite cover $\{V_\lambda \mid \lambda \in \Lambda\}$ of $\mathrm{im}(p_i)$ such that whenever $p_i(x), p_i(y) \in V_\lambda$ for some λ , we have $|\varphi(x) - \varphi(y)| < 1/n$. Now choose such a i_n and associated finite cover of $\mathrm{im}(p_{i_n})$ for each n . Then use the fact that $\mathcal{I}$ is $\aleph_1$ -cofiltered to take a cone $(j, f_n : j \rightarrow i_n)$ of the objects i_n . We then see that whenever we have $x, y \in X$ with $p_j(x) = p_j(y)$, then $\varphi(x) = \varphi(y)$. Thus, φ induces a well-defined function on the image of p_j . This function is continuous, since by construction for any $n \in \mathbb{N}_{>0}$ and $x \in X$, there is a neighborhood V of $p_j(x)$ such that whenever $p_j(y) \in V$ as

well, $|\varphi(y) - \varphi(x)| < 1/n$. By the Tietze extension theorem, this induced function can then be extended to a continuous function $\psi : X_j \rightarrow [0, 1]$. Then $\varphi = \psi \circ p_j$.

This shows the canonical map

$$\operatorname{colim}_{i \in \mathcal{I}^{\text{op}}} \operatorname{Hom}(X_i, [0, 1]) \rightarrow \operatorname{Hom}(\lim_{i \in \mathcal{I}} X_i, [0, 1])$$

is surjective.

Likewise, suppose we have $\alpha, \beta : X_i \rightrightarrows [0, 1]$ such that $\alpha \circ p_i = \beta \circ p_i$. For each $n \in \mathbb{N}_{>0}$, consider the set $D_n := \{x \in X_i \mid |\alpha(x) - \beta(x)| \geq 1/n\}$. Note that D_n is a closed subset of X_i , so it is compact. For any $x \in D_n$, we must have $x \notin \operatorname{im}(p_i)$. Using the contrapositive of lemma 1, we can conclude that there exists $f : j \rightarrow i$ such that $x \notin \operatorname{im}(X_f)$. Indeed, suppose not: then the slice category $\mathcal{I}/i$ is cofiltered, with the limit over this slice category being the same as the limit over $\mathcal{I}$. Also, by the contrary assumption, we have that $x \in \operatorname{im}(X_f)$ for any $f : j \rightarrow i$, so $X_f^{-1}(\{x\})$ is non-empty. Therefore, by lemma 1, the limit $p_i^{-1}(\{x\})$ would be non-empty, contradicting the assumption.

Now each $D_n \setminus \operatorname{im}(X_f)$ is open and we have just shown such sets cover D_n ; taking a finite subcover and then using the cofiltering assumption again, we conclude that there is a single $f_n : j_n \rightarrow i$ such that $\operatorname{im}(X_{f_n})$ is disjoint from D_n . Using the $\aleph_1$ -cofiltering assumption to take a cone of the f_n , we get that there is $f : k \rightarrow i$ such that $\operatorname{im}(X_f)$ is disjoint from all D_n . This implies that $\alpha \circ X_f = \beta \circ X_f$.

This shows the canonical map

$$\operatorname{colim}_{i \in \mathcal{I}^{\text{op}}} \operatorname{Hom}(X_i, [0, 1]) \rightarrow \operatorname{Hom}(\lim_{i \in \mathcal{I}} X_i, [0, 1])$$

is injective. □

Corollary 5. *The category **CompHaus** is locally $\aleph_1$ -copresentable.*

Proof. It suffices to show that the monad $S \mapsto \operatorname{Hom}_{\mathbf{CompHaus}}([0, 1]^S, [0, 1])$ is $\aleph_1$ -accessible. This monad is the composition of $[0, 1]^- : \mathbf{Set} \rightarrow \mathbf{CompHaus}^{\text{op}}$ followed by $\operatorname{Hom}_{\mathbf{CompHaus}}(-, [0, 1]) : \mathbf{CompHaus}^{\text{op}} \rightarrow \mathbf{Set}$. The first automatically preserves $\aleph_1$ -filtered colimits (and in fact all colimits) since it has a right adjoint. The second one preserves $\aleph_1$ -filtered colimits by the previous lemma.

Alternately, applying the general framework of Lawvere theories shows that $\mathbf{CompHaus}^{\text{op}}$ is equivalent to the category of functors $\mathcal{T} \rightarrow \mathbf{Set}$ preserving countable products, where $\mathcal{T}$ is the full subcategory of $\mathbf{CompHaus}$ of all spaces $[0, 1]^A$ where A is countable. Note that $\mathcal{T}$ is essentially small. We thus reproduce a result from [5] which also provides a nice description of a small set of generators of the operations of the $\aleph_0$ -ary algebraic theory. A more recent treatment in [7] refines this by providing a nice axiomatization of the relations of that theory. □

REFERENCES

- [1] J. Duskin. Variations on Beck's tripleability criterion. Reports of the Midwest Category Seminar III, pages 74–129. Springer Berlin Heidelberg, 1969.
- [2] P. Gabriel and F. Ulmer. Lokal präsentierbare Kategorien. *Lecture Notes in Mathematics*, 221, 1971.
- [3] Dirk Hoffman, Renato Neves, and Pedro Nora. Generating the algebraic theory of $C(X)$: The case of partially ordered compact spaces. *Theory and Applications of Categories*, 33(12):276–295, 2018. URL <http://www.tac.mta.ca/tac/volumes/33/12/33-12.pdf>.
- [4] M. Hušek and J. Rosický. Factorization and local presentability in topological and uniform spaces. *Topology and its Applications*, 259:251–266, 2019. ISSN 0166-8641. doi:<https://doi.org/10.1016/j.topol.2019.02.033>. URL <https://www.sciencedirect.com/science/article/pii/S0166864119300513>. William Wistar Comfort (1933-2016): In Memoriam.

- [5] J.R. Isbell. Generating the algebraic theory of $C(X)$. *Algebra Universalis*, 15 (2):153–155, 1982.
- [6] Saunders Mac Lane and Ieke Moerdijk. *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Universitext. Springer-Verlag New York, Inc., 1992. ISBN 978-0-387-97710-2. doi:10.1007/978-1-4612-0927-0. URL <https://link.springer.com/book/10.1007/978-1-4612-0927-0>.
- [7] Vincenzo Marra and Luca Reggiov. Stone duality above dimension zero: Axiomatising the algebraic theory of $C(X)$. *Advances in Mathematics*, 307:253–287, February 2017. ISSN 0001-8708. doi:10.1016/j.aim.2016.11.012. URL <http://dx.doi.org/10.1016/j.aim.2016.11.012>.
- [8] Vincenzo Marra and Luca Reggiov. A characterisation of the category of compact Hausdorff spaces, 2020. URL <https://arxiv.org/abs/1808.09738>.